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Yan

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(54) **HIGH CURRENT, MULTI-PHASE, SURFACE MOUNT INDUCTOR AND METHODS OF FABRICATION**

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H01F 27/06 (2006.01)
H01F 27/24 (2006.01)
H01F 41/02 (2006.01)

(52) **U.S. Cl.**
CPC **H01F 27/292** (2013.01); **H01F 27/06** (2013.01); **H01F 27/24** (2013.01); **H01F 41/0246** (2013.01); **H01F 2027/065** (2013.01)

(58) **Field of Classification Search**

CPC H01F 27/292; H01F 27/06; H01F 27/24; H01F 41/0246; H01F 2027/065; H01F 3/14; H01F 38/023; H01F 27/2852; H01F 27/306; H01F 336/20; H01F 336/233-234; H01F 336/212; H01F 336/192; H01F 336/83

See application file for complete search history.

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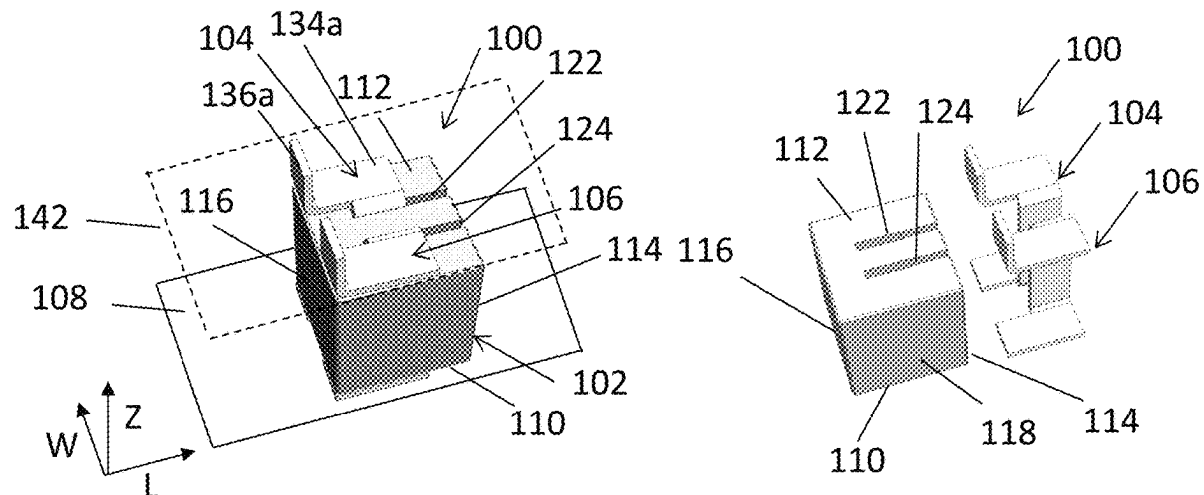
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(57) **ABSTRACT**

An integrated multi-phase inductor component assembly includes a single piece core or a multi-piece magnetic core that may be easily assembled with multiple coils arranged in a mirror-image relationship in the magnetic core. The coils include vertically extending, planar main winding sections extending in spaced apart but parallel vertical planes and horizontal sections extending in coplanar relationship on top and bottom sides of the magnetic core for surface mounting to a circuit board or through hole mounting to another circuit board or another component in power supply circuitry for an electronic device.

17 Claims, 6 Drawing Sheets



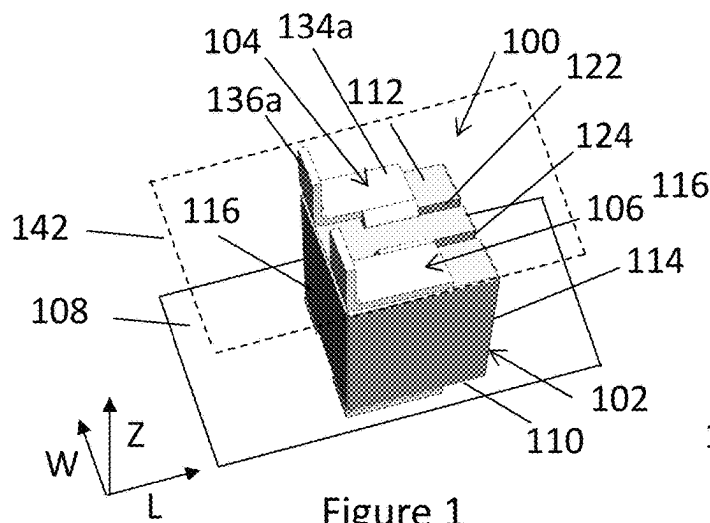


Figure 1

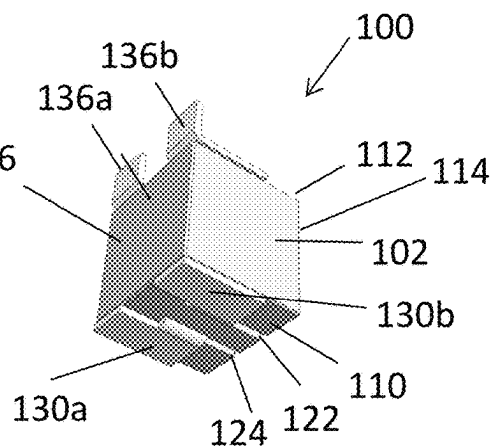


Figure 2

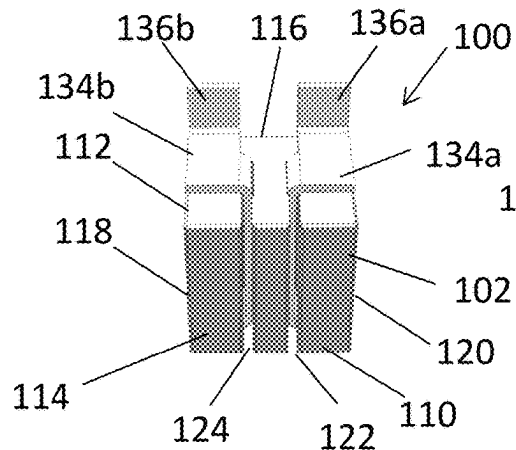


Figure 3

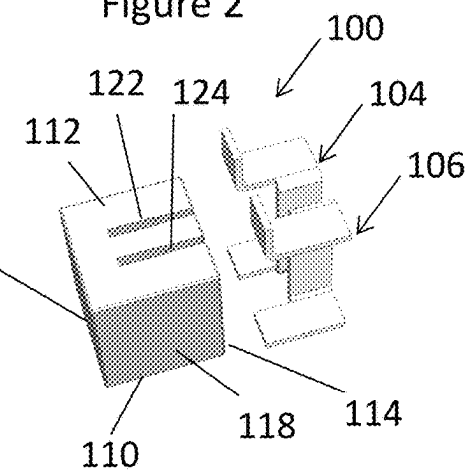


Figure 4

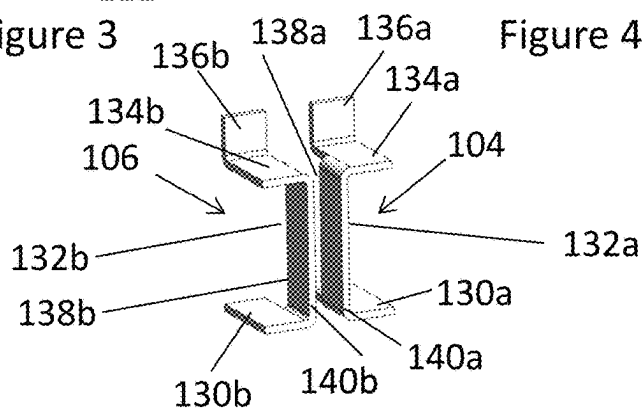


Figure 5

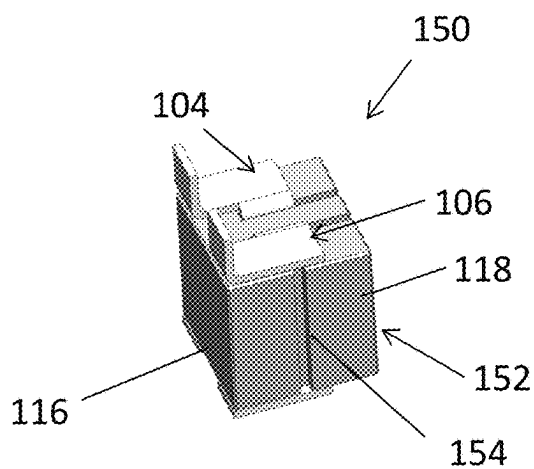


Figure 6

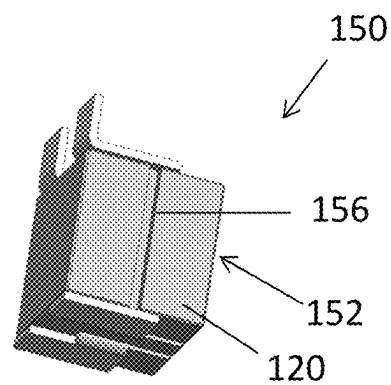


Figure 7

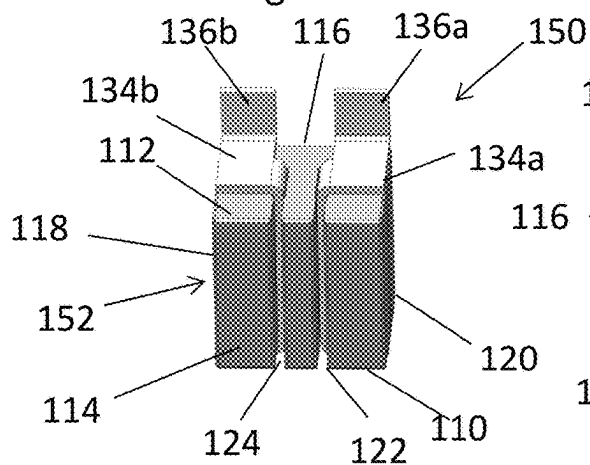


Figure 8

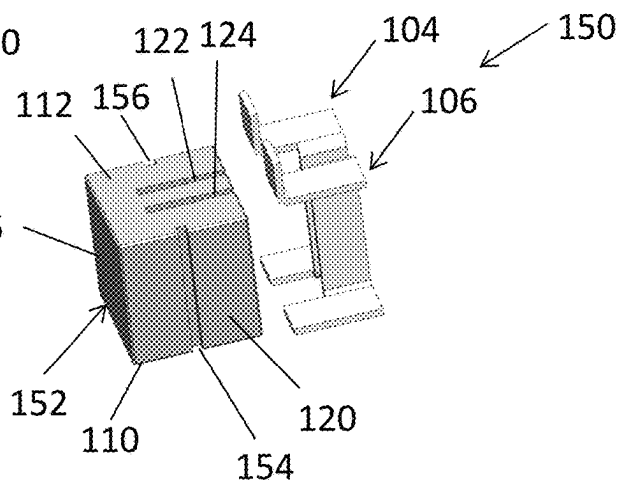


Figure 9

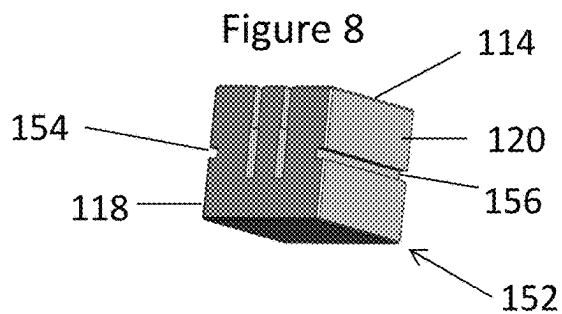


Figure 10

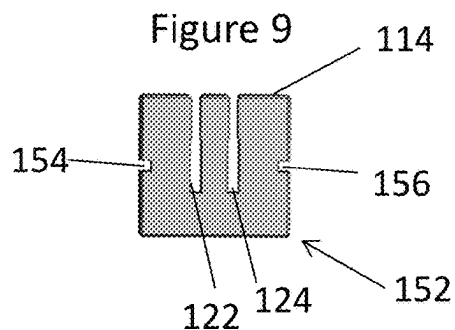


Figure 11

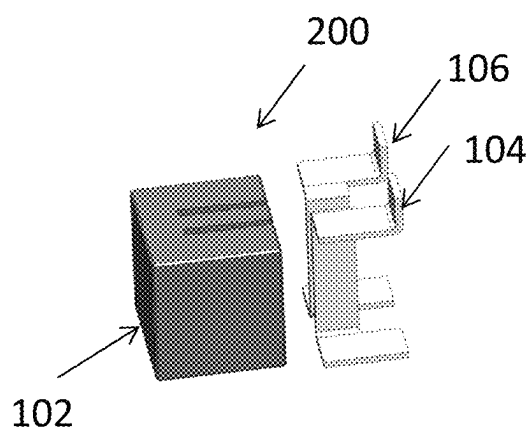


Figure 12

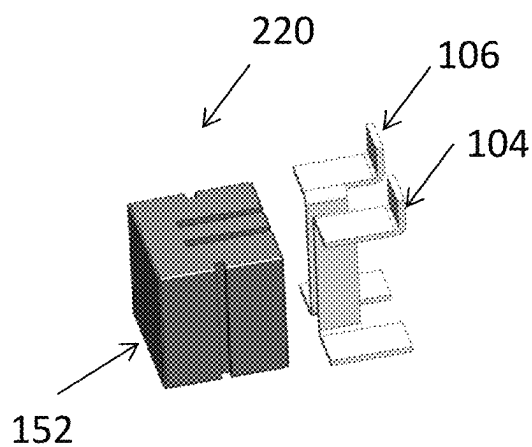


Figure 13

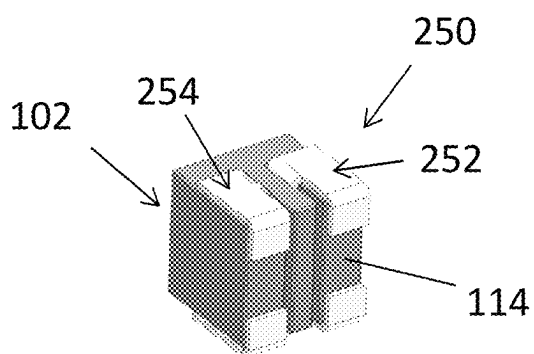


Figure 14

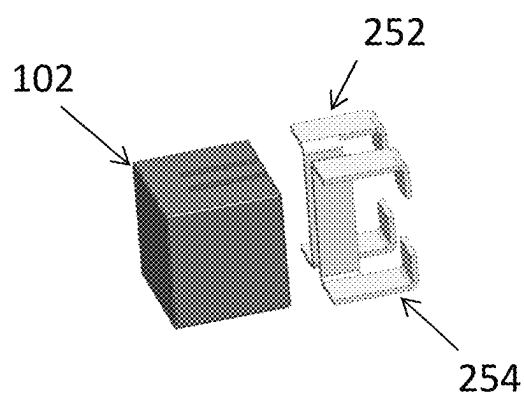


Figure 15

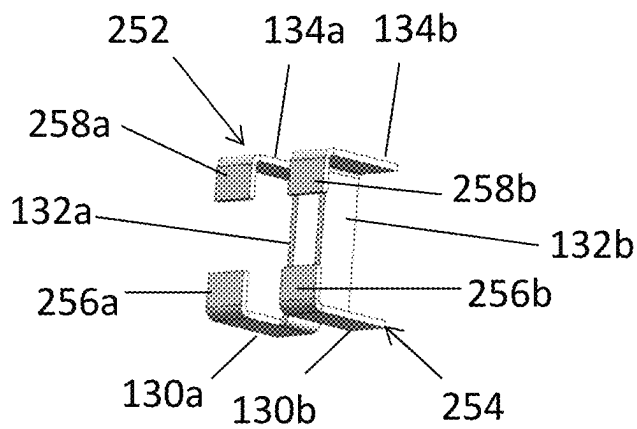


Figure 16

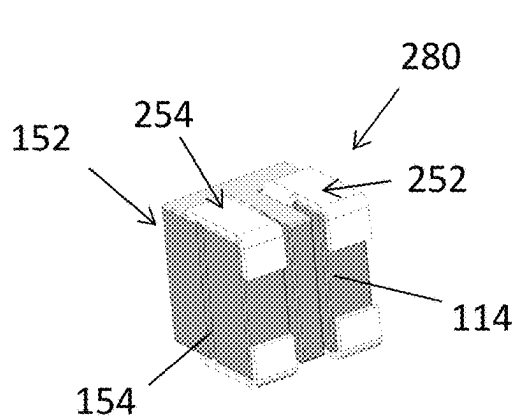


Figure 17

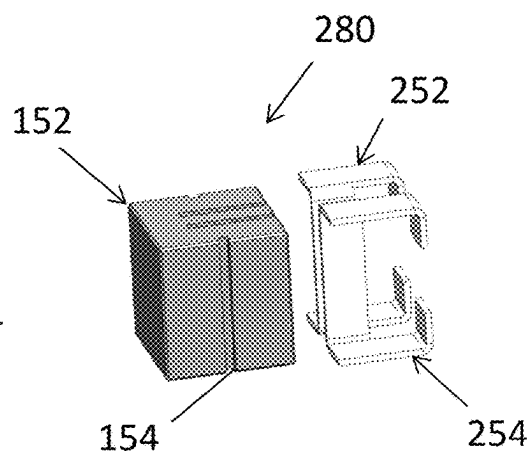


Figure 18

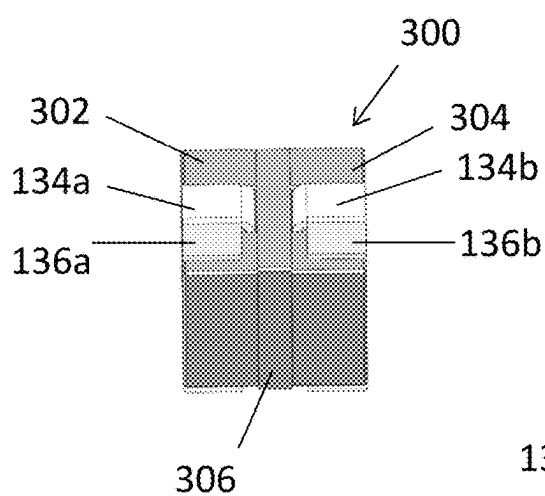


Figure 19

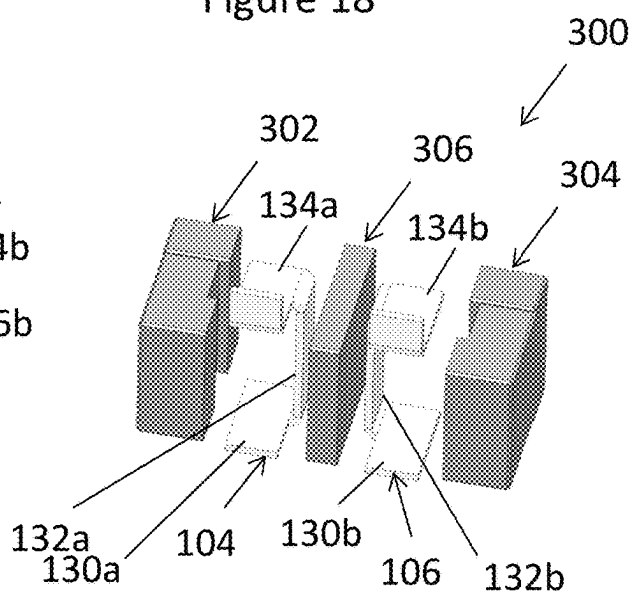


Figure 20

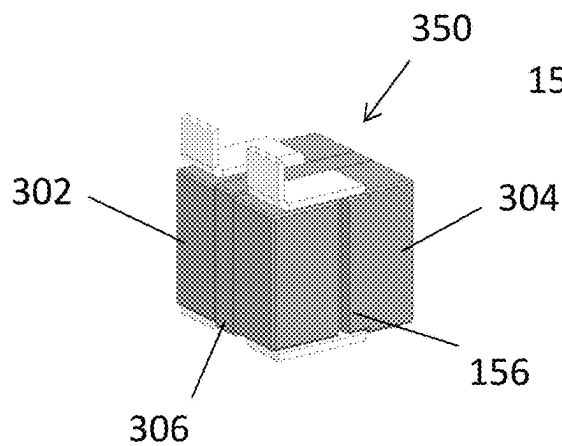


Figure 21

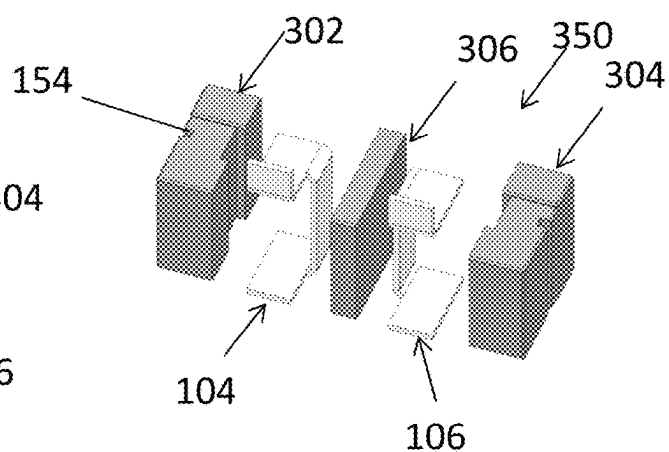


Figure 22

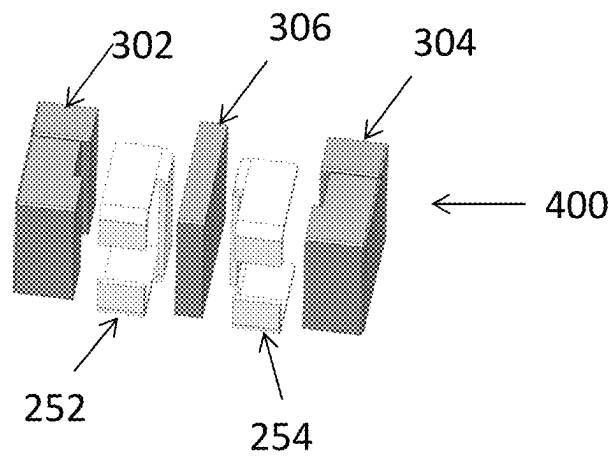


Figure 23

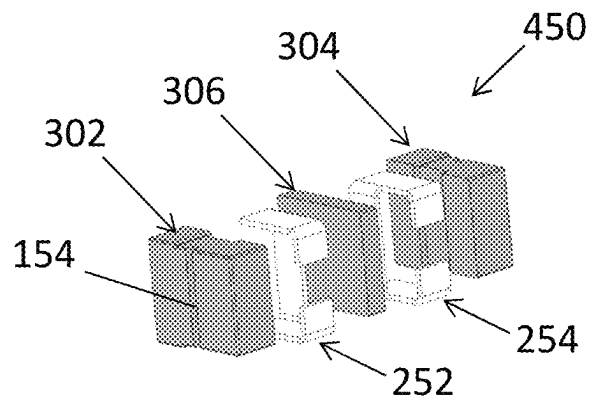


Figure 24

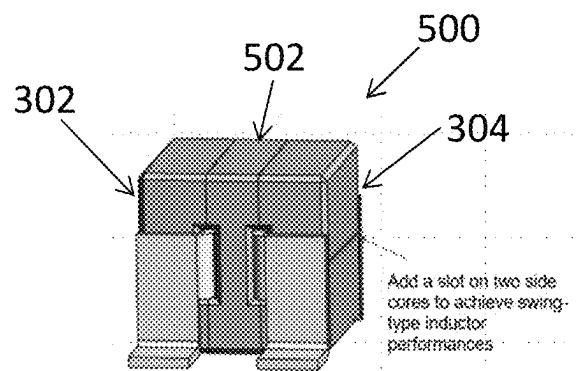


Figure 25

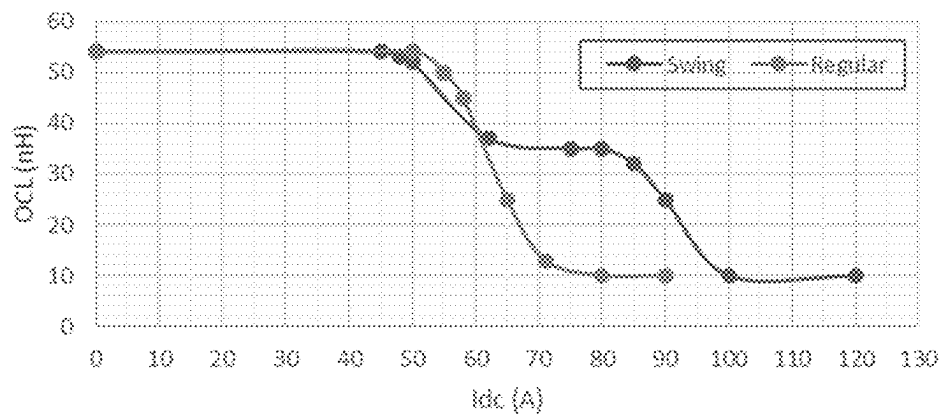


Figure 26

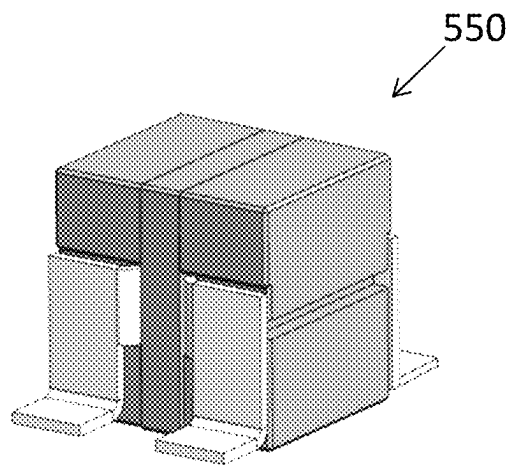


Figure 27

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HIGH CURRENT, MULTI-PHASE, SURFACE MOUNT INDUCTOR AND METHODS OF FABRICATION

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Application Ser. No. 63/139,192 filed Jan. 19, 2021, the complete disclosure of which is hereby incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

The field of the invention relates generally to surface mount electromagnetic component assemblies and methods of manufacturing the same, and more specifically to high current, multi-phase surface mount inductor components and methods of manufacturing the same.

Electromagnetic components such as inductors are known that utilize electric current and magnetic fields to provide a desired effect in an electrical circuit. Current flow through a conductor in the inductor component generates a magnetic field that can be concentrated in a magnetic core. The magnetic field, in turn, beneficially stores energy and releases energy, cancels undesirable signal components and noise in power lines and signal lines of electrical and electronic devices, or otherwise filters a signal to provide a desired output.

Increased power density in circuit board applications has resulted in a further demand for integrated multi-phase inductor solutions to provide power supplies in reduced package sizes. Such integrated multi-phase inductor components include a plurality of inductor coils provided on a common magnetic core structure. The coils may be magnetically coupled or non-coupled to realize different electromagnetic effects and performance characteristics. Relative to a number of discrete components each having an individual magnetic core and a single winding, such integrated multi-phase non-coupled and coupled inductor solutions may realize considerable space savings on a circuit board. Conventional integrated multi-phase non-coupled and coupled inductor solutions, however, are undesirably limited in some performance aspects and improvements are accordingly desired.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive embodiments are described with reference to the following Figures, wherein like reference numerals refer to like parts throughout the various drawings unless otherwise specified.

FIG. 1 is a top perspective view of a first exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 2 is a bottom perspective view of the component assembly shown in FIG. 1.

FIG. 3 is a front perspective view of the component assembly shown in FIGS. 1 and 2.

FIG. 4 is an exploded view of the of the component assembly shown in FIGS. 1, 2 and 3.

FIG. 5 is front perspective view of exemplary coils for the component assembly shown in FIGS. 1-4.

FIG. 6 is a top perspective view of a second exemplary embodiment of an integrated multi-phase inductor component assembly.

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FIG. 7 is a bottom perspective view of the component assembly shown in FIG. 6.

FIG. 8 is a front perspective view of the component assembly shown in FIGS. 6 and 7.

FIG. 9 is an exploded view of the component assembly shown in FIGS. 6, 7 and 8.

FIG. 10 is a perspective view of the magnetic core piece for the component assembly shown in FIGS. 6 through 9.

FIG. 11 is an end view of the magnetic core piece shown in FIG. 10.

FIG. 12 is an exploded view of a third exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 13 is an exploded view of a fourth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 14 is a front perspective view of a fifth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 15 is an exploded view of the component assembly shown in FIG. 14.

FIG. 16 is a perspective of conductive coils for the component assembly shown in FIGS. 14 and 15.

FIG. 17 is a perspective view of a sixth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 18 is an exploded view of the component assembly shown in FIG. 17.

FIG. 19 is a perspective view of a seventh exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 20 is an exploded view of the component assembly shown in FIG. 19.

FIG. 21 is a perspective view of an eighth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 22 is an exploded view of the component assembly shown in FIG. 20.

FIG. 23 is an exploded view of a ninth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 24 is an exploded view of a tenth exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 25 is a top view of an eleventh exemplary embodiment of an integrated multi-phase inductor component assembly.

FIG. 26 illustrates relative inductance characteristics of integrated multi-phase inductor component assembly according to embodiments of the present invention.

FIG. 27 is a top view of an eleventh embodiment of an integrated multi-phase inductor component assembly.

DETAILED DESCRIPTION OF THE INVENTION

State of the art telecommunications and computing (data-center, cloud, etc.) applications are requiring ever more powerful and high performance power supplies. In the case of medium and low power supplies (below 40 amps), a single-phase power supply architecture is adequate. However, with the latest processors, field programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), and cloud computing systems, higher levels of power and greater performance are in demand. New power supply modules for high current computing applications such as servers and the like are therefore needed.

In order to achieve new and higher thresholds of power delivery, multiphase power supply architectures are desired. Multiphase power supplies can be designed to be much more efficient than single-phase supplies at higher power levels, and the architecture also allows for more operational flexibility. Such flexibility could also include turning off some of the phases when they aren't needed to deliver the required power, and redundancy if failures occur in certain portions of the power supply system. Multiphase power supplies, however, require much more complex design strategies. Importantly, the increased complexity falls largely on the magnetic components of the power supplies. Innovative integrated inductor design, for both non-coupled and coupled inductors, is needed to address these challenges and enable a new standard in high performance power supplies for modern use cases.

For surface mount inductor component manufacturers, the challenge has been to provide increasingly miniaturized inductor components so as to minimize the area occupied on a circuit board by the inductor component (sometimes referred to as the component "footprint") and also its height measured in a direction perpendicular to a plane of the circuit board (sometimes referred to as the component "profile"). By decreasing the footprint and profile of inductor components, the size of the circuit board assemblies for electronic devices can be reduced and/or the component density on the circuit board(s) can be increased, which allows for reductions in size of the electronic device itself or increased capabilities of a device with a comparable size. Miniaturizing electronic components in a cost effective manner has, however, introduced a number of practical challenges to electronic component manufacturers in a highly competitive marketplace. Because of the high volume of inductor components needed for electronic devices in great demand, cost reduction in fabricating inductor components, without sacrificing performance, has been of great practical interest to electronic component manufacturers.

In general, each generation of electronic devices needs to be not only smaller, but offer increased functional features and capabilities. As a result, the electronic devices must be increasingly powerful devices. For some types of components, such as electromagnetic inductor components that, among other things, may provide energy storage and regulation capabilities, meeting increased power demands while continuing to reduce the size of inductor components that are already quite small, has proven challenging as a general proposition, and especially challenging for certain applications.

Multiple phase paralleled buck converters are widely utilized in power supply applications to manage higher current applications and provide enhanced capabilities and functions. A multiphase buck converter can more efficiently handle higher power than a single-phase buck converter of equivalent power output specification, imposing new demands for integrated multi-phase non-coupled and coupled inductors for power supply converter applications in telecommunications and computing applications due to their space saving advantages on a circuit board.

In some cases, the integrated multi-phase inductor components desirably operate with low inductance and high inductance for fast load transient response, high DC bias current resistance, and high efficiency individually. With continuous inductor size reduction, it is more and more challenging to achieve both high initial inductance and high DC bias current resistance together with conventional single step inductance drop characteristics.

Swing-type inductor components are known that are self-adjustable to achieve optimal trade-off between transient performance, DC bias current resistance and efficiency in power converter applications. Unlike other types of inductor components wherein the inductance of the component is generally fixed or constant despite the current load, swing-type inductor operate with an inductance that varies with the current load. Specifically, the swing-type inductor component may include a core that can be operated almost at magnetic saturation under certain current loads. The inductance of a swing core is at its maximum for a range of relatively small currents, and the inductance changes or swings to a lower value for another range of relatively higher currents. Swing-type inductors and their multiple step inductance rolloff characteristics can avoid the limitations of other types of inductor components in power converter applications, but are difficult to economically manufacture in desired footprints while still delivering desired performance. Improvements are accordingly desired.

Exemplary embodiments of integrated multi-phase inductor components are described hereinbelow that may more capably perform in higher current, higher power circuitry than conventional integrated multi-phase inductor components now in use. The exemplary embodiments of integrated multi-phase inductor component assemblies are further manufacturable at relatively low cost and with simplified fabrication processes and techniques. Further miniaturization of the exemplary embodiments of integrated multi-phase inductors is also facilitated to provide surface mount inductor components with smaller package size, yet improved capabilities in high current applications. Swing-type and non-swing type inductor components may be realized in an economical manner in desired package sizes with desired performance capabilities. Method aspects will be in part apparent and in part explicitly discussed in the description below.

FIGS. 1-4 illustrate various views of a first exemplary embodiment of an integrated multi-phase inductor component assembly **100**. The component **100** includes a magnetic core structure in the form of a single piece magnetic core **102** and a pair of conductive coils **104**, **106** (FIGS. 4 and 5) assembled thereto which may, in turn, be surface mounted to a circuit board **108** (FIG. 1). The circuit board **108** and the integrated multi-phase inductor component assembly **100** define a portion of power supply circuitry included in an electronic device. In a contemplated embodiment, the power supply circuitry on the circuit board **108** may implement a multiphase power supply architecture including a multiphase buck converter connected to the coils **104**, **106** of the integrated multi-phase inductor component assembly **100** in a high current computing application. As such multiphase power supply architecture and multiphase buck converter are known and within the purview of those in the art, further description thereof is omitted herein. The specific applications herein are, however, provided for the sake of illustration rather than limitation, and other applications are possible.

The single piece magnetic core **102** is formed with a bottom side **110** that abuts the circuit board **108** in use, a top side **112** opposite the bottom side **110**, opposing front and rear sides **114** and **116** interconnecting the top and bottom sides **110** and **112**, and left and right sides **118** and **120** interconnecting the top, bottom, front and rear sides **110**, **112**, **114** and **116**. The sides **110**, **112**, **114**, **116**, **118** and **120** in the core piece **102** in the example shown are generally orthogonally oriented to one another to form a box-like shape as shown. The sides **110**, **112**, **114**, **116**, **118** and **120**

have generally flat and planar outer surfaces, making the core piece **102** relatively easy to fabricate. For the purposes of the present description, the term “single” shall mean “only one” or “not one of several”. Thus, the single piece magnetic core **102** is distinguished from an assembly of a plurality of magnetic core pieces around the coils. The entire magnetic core structure is defined by the single piece core **102**, as opposed to a magnetic core structure otherwise defined by a combination of magnetic core pieces.

The magnetic core piece **102** has an overall length L along a first dimension extending parallel to the plane of the circuit board **108** (e.g., an x axis of a Cartesian coordinate system), a width W measured along a second dimension perpendicular to the first axis but still parallel to the plane of the circuit board **108** (e.g., a y axis of a Cartesian coordinate system), and a height H measured along a third dimension perpendicular to the first and second axis and the plane of the circuit board **108** (e.g., a z axis of a Cartesian coordinate system). In the illustrated examples, the width dimension may extend between the left and right sides **118**, **120**; the length dimension may extend between the front and rear sides **114**, **116**; and the height dimension may extend between the top and bottom sides **112**, **114**. As seen in the Figures, the length dimension, width dimension, and height dimension are nearly equal and the footprint of the component on the plane of the circuit board **108** is minimized, although other relative proportions of length, width and height of the magnetic core piece **102** are possible in other embodiments.

The front side **114** of the magnetic core piece **102** is further formed with coil slots **122** and **124** that respectively receive a portion of the coils **104**, **106**. The coil slots **122**, **124** extend generally straight and parallel to one another in a spaced apart relationship, extend completely from the bottom wall **110** to the top wall **112**, and extend only partly between the front side **114** and the rear wall **116**. The component assembly **100** including two coil slots **122**, **124** and two coils **104**, **106** in the common magnetic core piece **102** is well suited for a two-phase power system. The component assembly **100** is scalable however, to include additional numbers of coil slots and coils to easily adapt the component for further phases of multi-phase power applications or to obtain further space efficiencies by incorporating multiple coil windings on a common core structure. While specific geometry and location of the coil slots are shown and described, variations are of course possible with similar effect. For applications other than the aforementioned high power computing applications wherein single phase power supply architecture is acceptable, the component **100** could be provided with one coil slot and only one coil and therefore be a single-phase inductor component while still achieving at least some of the benefits described herein.

In contemplated embodiments, the magnetic core piece **102** may be fabricated utilizing soft magnetic particle materials and known techniques such as molding of granular magnetic particles to produce the desired shapes. Soft magnetic powder particles used to fabricate the magnetic core pieces may include Ferrite particles, Iron (Fe) particles, Sendust (Fe—Si—Al) particles, MPP (Ni—Mo—Fe) particles, HighFlux (Ni—Fe) particles, Megaflux (Fe—Si Alloy) particles, iron-based amorphous powder particles, cobalt-based amorphous powder particles, and other suitable materials known in the art. In some cases, magnetic powder particles may be coated with an insulating material such the magnetic core pieces may possess so-called distributed gap properties familiar to those in the art and fabricated in a known manner. The magnetic core pieces may be fabri-

cated from the same or different magnetic materials and as such may have the same or different magnetic properties as desired. The magnetic powder particles used to fabricate the magnetic core pieces may be obtained using known methods and techniques and molded into the desired shapes also using known techniques.

The coils **104**, **106** are substantially identically formed and shaped conductive elements that are arranged in an opposing, mirror image relationship to one another when assembled to the core piece **102**. Each coil **104**, **106** respectively includes, as best seen in FIG. 5: a first horizontally extending and generally planar surface mount terminal section **130a** and **130b**; a generally vertically extending and generally elongated, planar main winding section **132a**, **132b** extending perpendicularly to the respective surface mount terminal section **130a**, **130b**; a second horizontally extending and generally planar terminal section **134a** and **134b** extending from the planar main winding section **132a**, **132b** opposite the surface mount terminal section **130a** and **130b**; and terminal tabs **136a**, **136b** extending perpendicularly from the planar terminal section **134a** and **134b** in a direction upwardly and away from the sections **134a** and **134b**. The coils **104**, **106** each respectfully define less than one complete turn of an inductor winding in the magnetic core piece **102**, yet each of them has a sufficient thickness and cross sectional area to capably conduct higher current to meet performance requirements in higher power circuitry implemented on the circuit board **108**.

The coils **104**, **106** may be fabricated from a sheet of conductive material having a uniform thickness that is cut and formed or bent in the particular shape having the particular features shown. In contemplated embodiments the coils **104**, **106** may be preformed components in the assembly. That is, the formation of the coils **104**, **106** may be fully completed in advance and provided in the shape shown and described for assembly with the magnetic core piece **102** at a separate stage of manufacture. In the example shown, the horizontal sections **130a** and **130b** and **134a** and **134b** are seen in FIGS. 4 and 5 to have a different width than the vertical main winding section **132a** or **132b** such that a portion of the horizontal sections **130a**, **130b** and **134a**, **134b** protrudes beyond one of the side edges **138a**, **138b** of the main winding section **132a**, **132b** on one side thereof, while being generally flush with the other side edge **140a**, **140b** of the main winding section **132a**, **132b** in the other side. The larger width of the horizontal sections **130a**, **130b** and **134a**, **134b** relative to the main winding section **132a**, **132b** in each coil **104**, **106** imparts an asymmetry to the coils **104**, **106** along the vertical axis of the main winding section **132a**, **132b**. The presence of the terminal tabs **136a**, **136b** on the top side of the coils **104**, **106** but not on the bottom side of the coils **104**, **106** further imparts an asymmetry to the coils about a horizontal axis.

When assembled to the magnetic core piece **102**, the planar main winding sections **132a**, **132b** are received in the respective coil slots **122**, **124** of the magnetic core piece **102**. The coil slots **122**, **124** are slightly larger than the thickness of the coils **104**, **106** in the main winding section **132a**, **132b** to facilitate a slidable insertion of the main winding sections **132a**, **132b**. When the main winding sections **132a**, **132b** are fully seated in the coil slots **122**, **124** the surface mount terminal sections **130a**, **130b** underlie the bottom side **110** of the magnetic core piece **102** while the terminal sections **134a** and **134b** overlie the top side **112** of the magnetic core piece, and while the terminal tabs **136a**, **136b** are located adjacent the rear side **112**. The main winding sections **132a**, **132b** extend in spaced apart but parallel planes and are

relatively close together in the magnetic core piece **102**, while the horizontal sections **130a**, **130b** and **134a**, **134b** extend outwardly from the main sections **132a**, **132b** in a coplanar relationship on each of the bottom and top sides **110**, **112** of the magnetic core piece **102**. The horizontal sections **130**, **130b** and **134a**, **134b** respectively extend away from the respective coil slots **122**, **124** in a direction along-side and generally flush with the side walls **118**, **120** of the magnetic core piece. None of the horizontal sections **130a**, **130b**, **134a** and **134b** are in contact with or overlie the rear side **116** of the magnetic core piece **102**, however. In different embodiments the coils **104**, **106** may be magnetically coupled to one another inside the core piece **102**, or may be non-magnetically coupled by varying the spacing of the main winding sections **132a**, **132b** from one another. For the power converter application, the coils **104**, **106** are preferably non-magnetically coupled.

In another embodiment, the coils **104**, **106** need not be fully preformed as described above, but instead can be formed and bent into final shape after the main winding sections **132a**, **134** are inserted through the core piece **102** between the top and bottom sides **112**, **110** of the core piece **102**.

When fully assembled and secured in place on the magnetic core **102**, the surface mount terminal sections **130a**, **130b** on the bottom side **110** of the magnetic core piece **102** may be surface mounted to circuit traces on the circuit board **108** via known techniques such as soldering, and the terminal tabs **136a**, **136b** may be through hole mounted to a second circuit board or to another component (shown phantom as **142** in FIG. 1) via known techniques such as soldering. The second circuit board or component **142** is spaced apart from the circuit board **108**, and as such the terminal structure shown on the ends of the coils **104**, **106** establish one electrical connection the circuit board **108** and another to the second circuit board or component **142**, as opposed to completing two electrical connections on the same circuit board **108**. While exemplary surface mount and through-hole terminal geometry is shown and described to provide respective surface mount and through-hole mount structure on opposing sides of the magnetic core piece **102**, variations are of course possible. Also, while different types of mounting structure (i.e., surface mount and through-hole mount features) are shown and described on the opposing ends of the coils **104**, **106** in another embodiment the terminal mounting structure could be the same instead of different on each opposing end of the coils **104**, **106**. That is, and for example only, the coils **104**, **106** may alternatively be provided with through-hole terminal structure on both ends or with surface mount terminals on both ends. While one type of surface mount terminal and one type of through-hole terminal structure is shown and described, variations are, of course possible.

The single piece magnetic core piece **102** and the coils **104**, **106** may be separately fabricated in batch processing, and provided as preformed and prefabricated modular elements for assembly into components **100** in a reduced amount of time and at lower cost with respect to certain conventional integrated multi-phase inductor components including additional core pieces and/or coils that are more difficult to apply to the magnetic core.

FIG. 6-11 illustrate various views of a second exemplary embodiment of an integrated multi-phase inductor component assembly **150** that may likewise be surface mounted to the circuit board **108** (FIG. 1) on the bottom side as well as establish through-hole connection to the second circuit board or component **142** on the top side. The component **150**

includes a single piece magnetic core **152** and coils **104**, **106**. The single piece magnetic core piece **152** is similar to the single piece magnetic core **102** described above and as such similar manufacturing benefits are realized, but the single magnetic core piece **152** further includes physical gaps **154**, **156** respectively formed on an exterior of the sides **118**, **120** to realize further performance characteristics that are advantageous. In the example shown, the gaps **154**, **156** extend generally vertically in a straight or linear direction from the bottom side **110** to the top side **112** of the core piece **152** and are centered on the respective side walls **118**, **120** in the length dimension (i.e., along a direction perpendicular to front and rear sides **110**, **112**). As such, the gaps **154**, **156** generally oppose one another on the sides **118**, **120** of the magnetic core **152**. The gaps **154**, **156** are relatively simply shaped but provide an important performance characteristic for certain applications, namely swing-type inductor functionality as a result of the reduced cross-sectional area of the core including the gaps **154**, **156** relative to the component **100** without the gaps **154**, **156**. While specific geometry and location of the gaps **154**, **156** are shown and described, variations are possible with similar effect to produce different degrees of swing-type functionality.

FIG. 26 illustrates exemplary inductance characteristics of the integrated multi-phase inductor component assemblies **100** and **150** relative to one another. The inductance characteristics are shown in the form of inductance plots wherein inductance values correspond to the vertical axis and wherein current values correspond to the horizontal axis. The inductance plot for the component **100** is shown as “regular” while the inductance plot for the component **150** is shown as “swing” in FIG. 26.

As seen in the inductance plots of FIG. 26, the inductor component assembly **100** of the “regular” component **100** exhibits a fixed and generally constant inductance value indicated by the horizontal line at the left-hand side of FIG. 26 that represents a constant open circuit inductance (OCL) value over a normal operating range of current values. That is, the open circuit inductance (OCL) value of the component **100** is the same regardless of the actual current load in use within the normal operating range of the inductor component assembly **100**. The “swing” component **150** exhibits the same OCL value over about the same current range as the component **100**.

As also seen in FIG. 26, and represented by the sloped portion of the plot for the “regular” component **100**, when the inductor component assembly **100** is operated at a current up to its saturation current (I_{sat}) that represents a full load inductance (FLL) or full load operation, the inductor component assembly **100** exhibits a fixed and generally constant inductance value corresponding to a full load inductance (FLL) value regardless of the actual current load. In contrast, and as can be seen in the other plot in FIG. 26 for the “swing” component **150**, the swing component has an inductance that varies with the current load, and specifically can be operated almost at magnetic saturation under certain current loads, while changing or swinging to a lower inductance value for another range of relatively higher currents. As such, the “swing” inductor **150** exhibits multiple steps of inductance rolloff characteristics while the “regular” conductor does not (i.e., the “regular” conductor has a single step rolloff characteristic). The multiple step rolloff characteristics of the swing inductor **150** provide substantial performance benefits for certain power converter applications relative to the regular inductor **100** (i.e., a non-swing-type inductor). Specifically, the swing inductor may operate with high inductance at a range of light (i.e., lower) current loads

until eventually becoming saturated via a smaller cross sectional of the core (via the additional gaps such as the gaps **154** and **156** described above) until the OCL drops and realizes a higher DC bias resistance for a range of heavy (i.e. higher) current loads, while returning back to the high inductance when the current load returns back to the range of light current load.

Table 1 below shows exemplary length, width and height dimensions and expected performance parameters of components **100** and **150**, again with the component **100** indicated as “regular” and with the component **150** indicated as “swing” in the inductor type column. The tabulated performance parameters are well suited for a power converter application but are difficult to achieve in conventional inductor component constructions in similar package sizes.

TABLE 1

No.	L (mm)	W (mm)	H (mm)	OCL +/- 10% (nH)	Isat 1 @100 C. (A)	Isat2 @100 C. (A)	FLL 1 @Isat1 Typ. (nH)	FLL2 @Isat2 Min. (nH)	DCR @20 C. (mOhm)	Irms For 40 C. ΔT (A)	Inductor Type
1	6.7	6.7	5.15	54	50	90	35	25	0.145	43	Swing
2	6.7	6.7	6.15	67	50	90	43	31	0.173	40	Swing
3	6.7	6.7	7.15	79	50	90	51	37	0.201	37	Swing
4	6.7	6.7	8.15	91	50	90	59	43	0.230	35	Swing
5	6.7	6.7	5.15	54	58	90	39	10	0.145	43	Regular
6	6.7	6.7	6.15	67	58	90	48	10	0.173	40	Regular
7	6.7	6.7	7.15	79	58	90	57	10	0.201	37	Regular
8	6.7	6.7	8.15	91	58	90	66	10	0.230	35	Regular

It is noted that in some embodiments, the gaps **154**, **156** can be utilized, or not utilized, on a per-phase basis in the integrated multi-phase inductor component. That is, one phase may be operated with “regular” inductance while another phase may be operated with “swing” inductance when the component is mounted to the circuit board **108**.

FIGS. **12** and **13** are exploded views of further embodiments of integrated multi-phase inductor component assemblies **200** and **220** that may likewise be surface mounted to the circuit board **108** (FIG. **1**) on the bottom side as well as establish through-hole connection to the second circuit board or component **142** on the top side. The inductor component assemblies **200** and **220** are respectively similar to the components **100** and **150** described above and realize similar performance and manufacturing benefits. Comparing FIG. **12** to FIG. **4** and comparing FIG. **13** to FIG. **9** it is seen that the coils **104** and **106** are oriented differently in the components **200** and **220** relative to the components **100** and **150**. Specifically, the combination of coils **104** and **106** are assembled to the core piece **102** or **152** after being oriented in a 180° orientation from that shown in FIGS. **4** and **9**. As a result the horizontal sections **130a**, **130b**, **134a**, **134b** of the coils **104**, **106** extend in opposite directions in the completed components **200** and **220** (i.e., toward the front side **114** instead of the rear wall **116**) on the bottom and top sides of the magnetic core piece **102** in the component assembly **200** or in the component assembly **220**.

FIGS. **14-16** are various views an embodiment of an integrated multi-phase inductor component assembly **250** that may likewise be surface mounted to the circuit board **108** (FIG. **1**) on the bottom side as well as establish surface mount connection to the second circuit board or component **142** on the top side as described below. The inductor component assembly **200** provides some of the same benefits of the components described above but with a different structure of the coils. The component assembly **250** includes

the core piece **102** and coils **252**, **254** that each respectively include horizontal sections **130**, **130b**, **134a**, **134b** and the vertical main winding sections **132a**, **132b**. The distal ends of the horizontal sections **130**, **130b**, **134a**, **134b** further include terminal sections **256a**, **256b** and **258a**, **258b** that extend perpendicularly to the horizontal sections **130a**, **130b**, **134a**, **134b** and in a direction toward one another. That is, the sections **256a**, **256b** and **258a**, **258b** extend vertically at a distance from the main winding sections **132a**, **132b**. The sections **256a**, **256b**, **258a**, **258b** extend generally coplanar to one another but in a plane extending perpendicular to the plane of either the main winding section **132a** or the main section **132b**.

When the main winding sections **132a**, **132a** are assembled to the core piece **102** as described above, the

sections **256a**, **256b** and **258a**, **258b** wrap around and overlie the front side **114** of the magnetic core piece. As such, the component **250** may be surface mounted to the circuit board **108** via the sections **256a**, **256b** and **258a**, **258b** on the front side **114** of the core piece, or the component **250** may be surface mounted to the circuit board **108** via the sections **130a**, **130b**, **134a**, **134b** on the top and bottom side **112** and **110**. The manufacturing benefits are otherwise similar to the previous embodiments.

FIGS. **17** and **18** are different views of an embodiment of an integrated multi-phase inductor component assembly **280** that may likewise be surface mounted to the circuit board **108** (FIG. **1**) on the bottom side as well as establish surface mount connection to the second circuit board or component **142** on the top side. The component assembly **280** is similar to the component **250** but includes the core piece **152** instead of the core piece **102**. By virtue of the core piece **152**, swing-type functionality and performance is realized, while the manufacturing and assembly advantages of the components **280** and **250** are similar.

FIGS. **19** and **20** are different views of another embodiment of an integrated multi-phase inductor component assembly **300** that may likewise be surface mounted to the circuit board **108** (FIG. **1**) on the bottom side as well as establish surface mount connection to the second circuit board or component **142** on the top side. The inductor component assembly **300** includes a multiple-piece magnetic core construction instead of the single piece magnetic core constructions described above in the previous embodiments. As such, the component assembly **300** includes a first core piece **302** and a second magnetic core piece **304** that opposes the first core piece, with a third magnetic core piece **306** in between. Coils **104**, **106** are respectively located between the core piece **302** and the core piece **306** and between the core piece **306** and the core piece **304**.

As shown in the illustrated example, interior walls of the core pieces **302** and **304** include vertically extending chan-

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nels that receive the main winding sections **132a**, **132b** of the coils **104**, **106** in the assembly **300**. The top side of the magnetic core pieces **302** and **304** further includes a recessed surface receiving the horizontal section **134a** or **134b** as shown in the Figures. A length of the horizontal section **134a**, **134b** in each coil **104**, **106** is reduced relative to a length of the horizontal sections **130a**, **130b**. As such, the horizontal sections **130a**, **130b** extend flush with the end of the magnetic core in the view of FIG. 19, while the horizontal sections **134a**, **134b** are spaced from the end edge of the magnetic core, resulting in the terminal tabs **136a**, **136b** being flush with the opposing sides of the magnetic core while being spaced from the third side.

The multi-piece core construction of the component assembly **300** lends itself to modular expansion to add core pieces and coils to scale the number of inductors in the resultant integrated multi-phase core components with relative ease from a small set of component parts, whereas the single piece core constructions require an inventory of different core pieces to provide components with different numbers of coils. The multi-piece core construction may also include core pieces having different magnetic materials for still further performance variations to meet the needs of particular applications.

FIGS. 21 and 22 are different views of another embodiment of an integrated multi-phase inductor component assembly **350** that may likewise be surface mounted to the circuit board **108** (FIG. 1) as well as connected to the second circuit board or component **142** on the top side. The component assembly **300** is similar to the component **300**, with the addition of physical gaps **154**, **156** respectively formed in the magnetic pieces **302**, **304** to realize swing-type inductor performance characteristics.

FIG. 23 is an exploded view of another embodiment of an integrated multi-phase inductor component assembly **400** that may likewise be surface mounted to the circuit board **108** (FIG. 1) on the bottom side as well as establish surface mount connection to the second circuit board or component **142** on the top side. The component assembly **400** is similar to the assembly **300** but includes coils **252**, **254** instead of the coils **104**, **106**.

FIG. 24 is an exploded view of another embodiment of an integrated multi-phase inductor component assembly **450** that may likewise be surface mounted to the circuit board **108** (FIG. 1) on the bottom side as well as establish surface mount connection to the second circuit board or component **142** on the top side as described below. The component assembly **450** is similar to the component **400** but includes physical gaps **154**, **156** respectively formed in the magnetic pieces **302**, **304** to realize swing-type inductor performance characteristics.

FIG. 25 is a top perspective view of another embodiment of an integrated multi-phase inductor component assembly **500** that may likewise be mounted to the circuit board **108** (FIG. 1) on the bottom side as well as connected to the second circuit board or component **142** on the top side. The component assembly **500** includes the core pieces **302** and **304** and coils similar to those described above. A third core piece **502** extends between the coils and the core pieces **302** and **304**, which unlike the core piece **306** in the previous embodiments, is formed with coil channels on the opposing sides thereof. As such, the main winding sections **132a**, **132b** of the coils in the component assembly **500** are received partly in the coil channels of the core pieces **302**, **304** and partly in the coil channels of the third core piece **510**. Physical gaps similar to the gaps **154**, **156** may optionally be

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formed in the core pieces **302**, **304** to provide desired swing-type inductor performance characteristics.

FIG. 27 is a top perspective view of another embodiment of an integrated multi-phase inductor component assembly **550** that may likewise be mounted to the circuit board **108** (FIG. 1) on the bottom side as well as connected to the second circuit board or component **142** on the top side. The inductor component assembly **550** is similar to the component **350** described above but includes the same terminal structure on each opposing side of the magnetic core, whereas the coils **104**, **106** in the component assembly **350**, for example, include different terminal structure on each opposing side of the magnetic core. In the example shown, the terminal structure on the opposing sides of magnetic core structure includes planar terminal sections with terminal tabs extending therefrom as described above. As such the component assembly **500** may be through-hole mounted to the circuit board **108** and through-hole mounted to the second circuit board or component **252**.

The various component assemblies described above offer a considerable variety of integrated multi-phase inductors with or without swing-type inductor functionality while using a small number of component parts that are manufacturable to provide small components at relatively low cost with superior performance advantages. Particularly in the case of high power density electrical power system applications such as the multi-phase power supply circuits and power converters for computer servers, computer workstations and telecommunication equipment, the swing-type inductors components described herein are operable with desired package size and desired efficiency that is generally beyond the capability of conventionally constructed surface mount swing-type inductor components.

The benefits and advantages of the inventive concepts disclosed are now believed to be evident in view of the exemplary embodiments disclosed.

An inductor component assembly for a circuit board has been disclosed, the inductor component assembly including a magnetic core structure having a top side and a bottom side opposing the top side, and at least one conductive coil received in a portion of the magnetic core structure. The at least one conductive coil includes a first terminal structure extending on the bottom side for connection to the circuit board, and a second terminal structure on the top side.

Optionally, the second terminal structure may be configured for through-hole mounting to another circuit board or electrical component. The second terminal structure may include a first planar terminal section and a terminal tab extending perpendicular to the first planar terminal section. The magnetic core structure may also include opposing front and rear sides extending between the top and bottom side, wherein the terminal tab extends alongside the front wall or the rear wall without extending over the front wall or rear wall.

Also optionally, the magnetic core structure may include opposing front and rear sides, and at least one of the front wall or rear wall may include at least one coil slot. The at least one coil slot may extend fully between the top and bottom sides. The at least one coil slot also may extend only partly between the front and rear sides. The magnetic core structure may include opposing left and right sides interconnecting the front and rear sides, and the opposing left and right sides may include at least one physical gap producing multiple step inductance rolloff characteristics in operation of the component.

As further options, the second terminal structure may be configured for surface mounting to another circuit board or

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electrical component. The magnetic core structure may also include opposing front and rear sides extending between the top and bottom side, wherein the first terminal structure and the second terminal structure each wrap around one of the front and rear sides to provide surface mount terminal structure on the front side. One of the front wall or the rear wall may include at least one coil slot. The at least one coil slot may extend fully between the top and bottom sides. The at least one coil slot may extend only partly between the front and rear sides. The magnetic core structure may include opposing left and right sides interconnecting the front and rear sides, wherein the opposing left and right sides includes at least one physical gap producing multiple step inductance rolloff characteristics in operation of the component.

The magnetic core structure optionally may be entirely defined by a single magnetic core piece. The single magnetic core piece may have a length dimension and width dimension measured parallel to a plane of the circuit board, wherein the length dimension and the width dimension are substantially equal to one another.

Alternatively, the magnetic core structure may optionally be defined by multiple magnetic core pieces. The multiple magnetic core pieces may include at least first and second identically shaped magnetic core pieces arranged in a mirror image relationship to one another, and a third magnetic core piece extending between the first and second magnetic core pieces. At least one the first and second magnetic core pieces may include a physical gap producing multiple step inductance rolloff characteristics in operation of the component.

The first terminal structure and the second terminal may respectively define a surface mount terminal structure, a through-hole terminal structure or a combination of a surface mount terminal structure and a through-hole terminal structure.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. An inductor component assembly for a circuit board, wherein the inductor component assembly comprising:

a magnetic core structure having a top side and a bottom side opposing the top side and at least one coil slot extending from the bottom side to the top side; and at least one conductive coil comprising a planar surface mount terminal section extending on the bottom side for connection to the circuit board, a planar terminal section on the top side, a terminal tab extending perpendicular to the planar terminal section, and a planar main winding section extending perpendicularly to the planar surface mount terminal section;

wherein the at least one conductive coil is received in the magnetic core structure and the planar main winding section extends through the at least one coil slot.

2. The inductor component assembly of claim 1, wherein the terminal tab is configured for through-hole mounting to another circuit board or electrical component.

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3. The inductor component assembly of claim 1, wherein the magnetic core structure further includes opposing front and rear walls extending between the top and bottom side, the opposing front and rear walls comprising a front wall and a rear wall, wherein the terminal tab extends alongside the front wall or the rear wall without extending over the front wall or rear wall.

4. The inductor component assembly of claim 1, wherein the magnetic core structure further includes opposing front and rear sides, and at least one of the front side or wall side including the at least one coil slot.

5. The inductor component assembly of claim 4, wherein the at least one coil slot extends only partly between the front and rear sides.

6. The inductor component assembly of claim 4, wherein the magnetic core structure includes opposing left and right sides interconnecting the front and rear sides, and wherein the opposing left and right sides includes at least one physical gap producing multiple step inductance rolloff characteristics in operation of the component.

7. The inductor component assembly of claim 1, wherein the planar surface mount terminal section is configured for surface mounting to another circuit board or electrical component.

8. The inductor component assembly of claim 7, the planar surface mount terminal section further comprising a terminal section extending perpendicular to the planar surface mount terminal section, wherein the magnetic core structure further includes opposing front and rear sides extending between the top and bottom side, wherein the terminal section wraps around one of the front and rear sides to provide surface mount terminal structure on the front side.

9. The inductor component assembly of claim 7, wherein the magnetic core structure further includes a front wall and a rear wall, and wherein one of the front wall or the rear wall includes the at least one coil slot.

10. The inductor component assembly of claim 9, wherein the at least one coil slot extends fully between the top and bottom sides.

11. The inductor component assembly of claim 9, wherein the at least one coil slot extends only partly between the front and rear walls.

12. The inductor component assembly of claim 7, wherein the magnetic core structure includes opposing left and right sides interconnecting front and rear sides of the magnetic core structure, and wherein the opposing left and right sides includes at least one physical gap producing multiple step inductance rolloff characteristics in operation of the component.

13. The inductor component assembly of claim 1, wherein the magnetic core structure is entirely defined by a single magnetic core piece.

14. The inductor component assembly of claim 13, wherein the single magnetic core piece has a length dimension and width dimension measured parallel to a plane of the circuit board, and wherein the length dimension and the width dimension are substantially equal to one another.

15. The inductor component assembly of claim 1, wherein the magnetic core structure is defined by multiple magnetic core pieces.

16. The inductor component assembly of claim 15, wherein the multiple magnetic core pieces includes at least first and second identically shaped magnetic core pieces arranged in a mirror image relationship to one another, and a third magnetic core piece extending between the first and second magnetic core pieces.

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17. The inductor component assembly of claim **16**, wherein at least one the first and second magnetic core pieces includes a physical gap producing multiple step inductance rolloff characteristics in operation of the component.

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